



DEPARTAMENTO
DE INFORMÁTICA
PUC-RIO

Properties of agent systems

Perspectives on Agents and Environments



Outline

- **What is an agent?**
- Intelligent agents
- Agent properties
- Environment properties
- Example



What is an Agent?

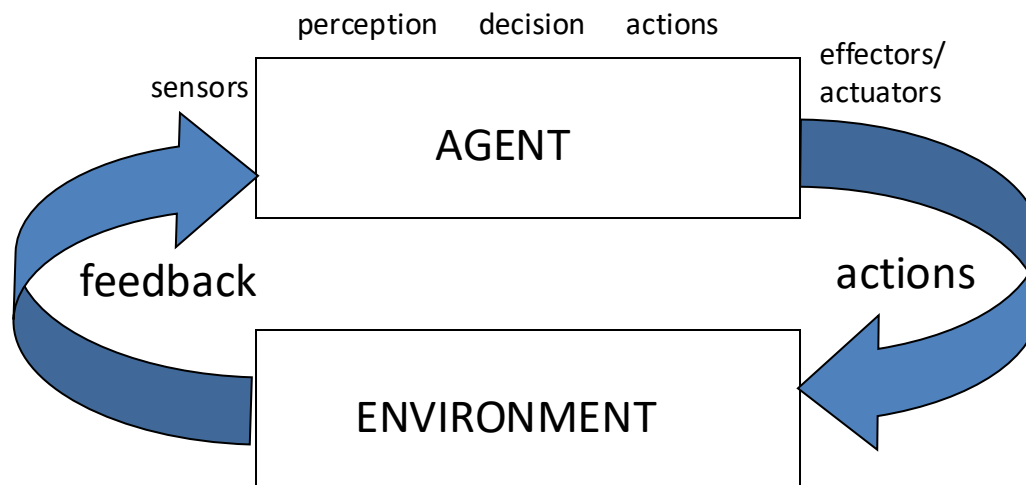
- *Autonomy* is a key aspect within an agent
 - In other words, an agent is capable of *acting independently*
- Thus: *an agent is a computer system capable of autonomous action in some environment in order to meet its design objectives*



What is an Agent?

- We think of an agent as being in a closed-couple continual interaction with its environment:

sense – decide – act – sense – decide – act...



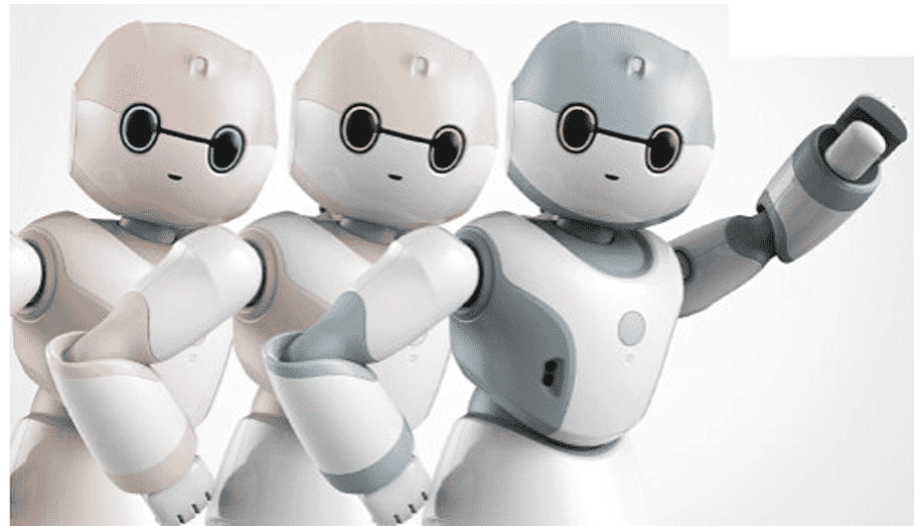
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Intelligent Agents

- Intelligent agents typically exhibit the following types of behavior:
 - *reactive*
 - *proactive*
 - *social*



Reactive

- The real world is constantly changing!
 - Environment is changing
- Many (most?) interesting environments are *dynamic*



Reactive

- A *reactive* system is one that maintains an *ongoing interaction with its environment*
- *Responds to changes* (in time for the response to be useful)

Proactive

- Reacting to an environment is relatively easy to implement
 - E.g., stimulus → response rules
- But we generally want agents to *do things for us*
 - Hence *goal directed behavior*



Proactive

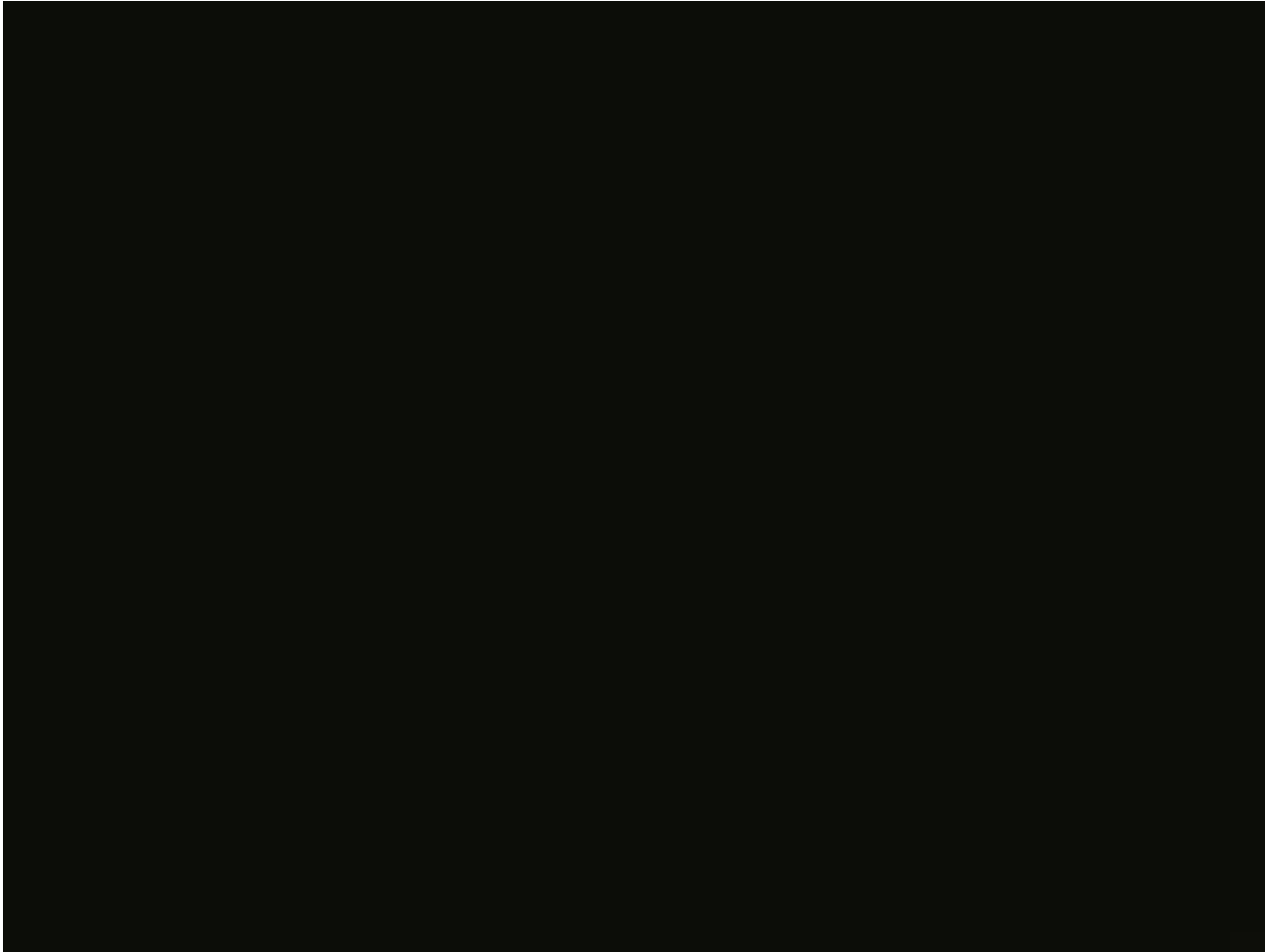
- Pro-activeness = generating and attempting to achieve goals
 - not driven solely by events
 - taking the initiative
 - Recognizing opportunities



Balancing Reactive and Goal-Oriented Behavior

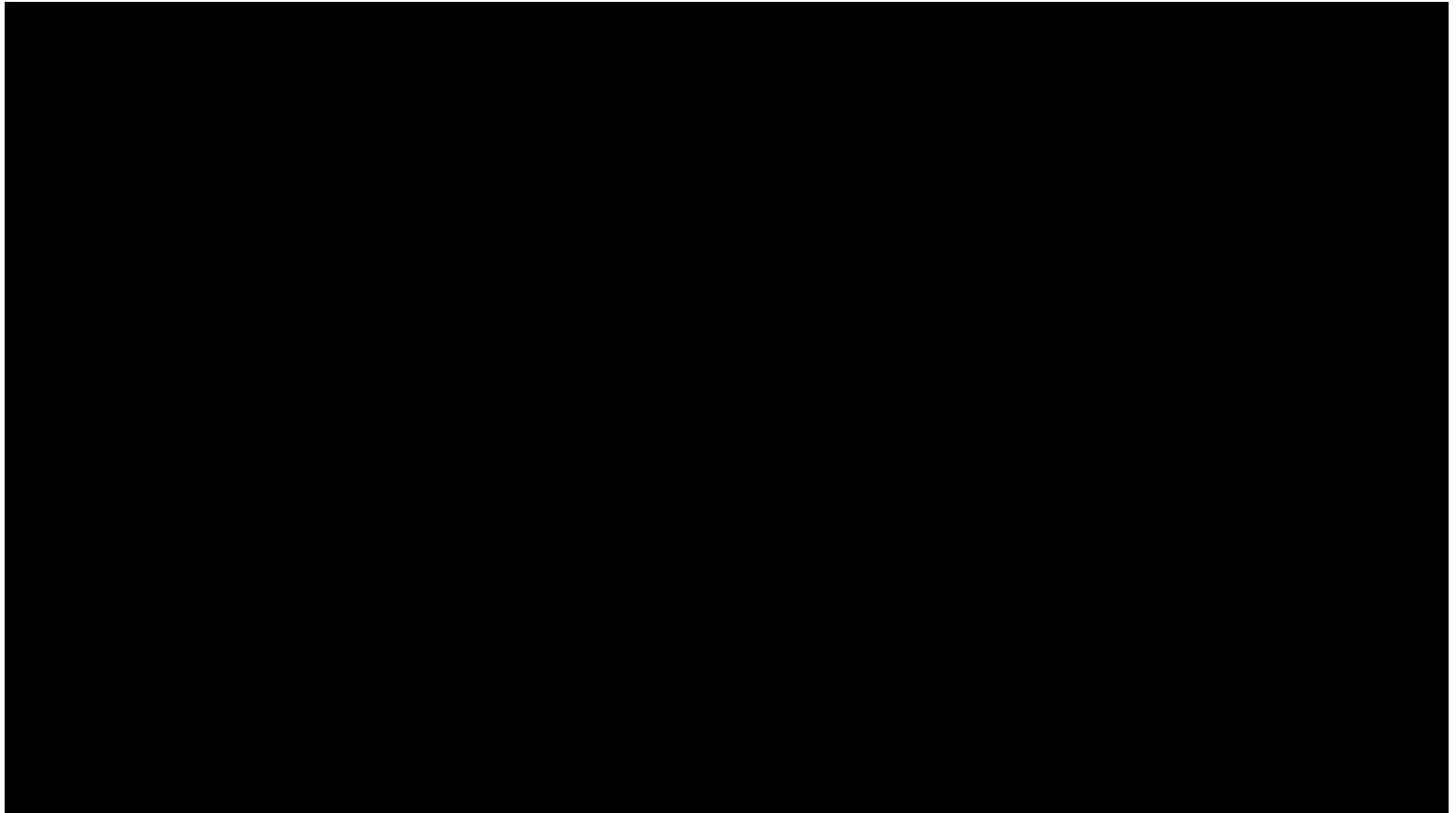
- We want our *agents to be reactive*
- We want our *agents to systematically work towards long-term goals*
- However, these *two considerations can be at odds with one another*

Stanley in action



<http://www.youtube.com/watch?v=M2AcMnfzpNg>

Since then...



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Agent Properties

- The real world is a *multiagent* environment
- Agents cannot go around attempting to achieve goals *without taking other agents into account*



Agent Properties

- Some *goals can only be achieved with cooperation, coordination or negotiation*



Agent Properties

- *Social ability* = *cooperation, coordination, and negotiation*



Agent Properties

- *Cooperation* is working together as a team to achieve a shared goal
- Typically needed when:
 - No agent can achieve the goal alone
 - Cooperation obtains a better result

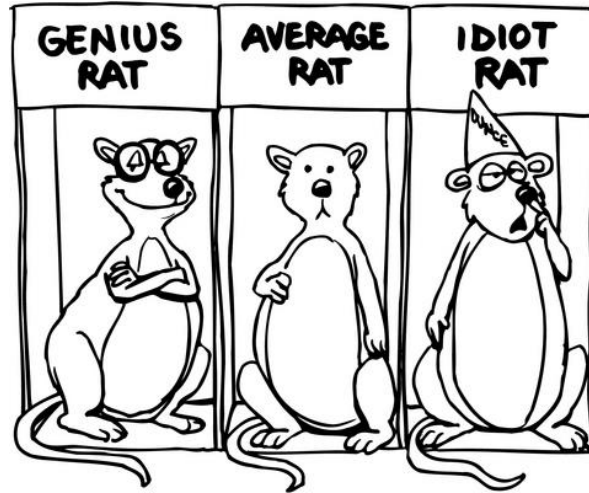
Agent Properties

- *Coordination* is managing the interdependencies between activities
- For example:
 - A non-sharable resource in the environment
 - Agents need to coordinate to use this resource

Agent Properties

- *Negotiation* is the ability to reach agreements on matter of common interest
- For example:
 - You have a TV in your house
 - You want to watch a movie
 - Your roommate wants to watch football
 - Deal: He watches football tonight and you watch a movie tomorrow

Other Agent Properties



- Are there other properties that characterize agents?
 - Autonomy
 - Adaptivity
 - Rationality
 - Mobility
 - Etc.

Agent Properties

Autonomy

- agent's *ability to act independently* and thus determine how to achieve its delegated goals/tasks



Adaptivity

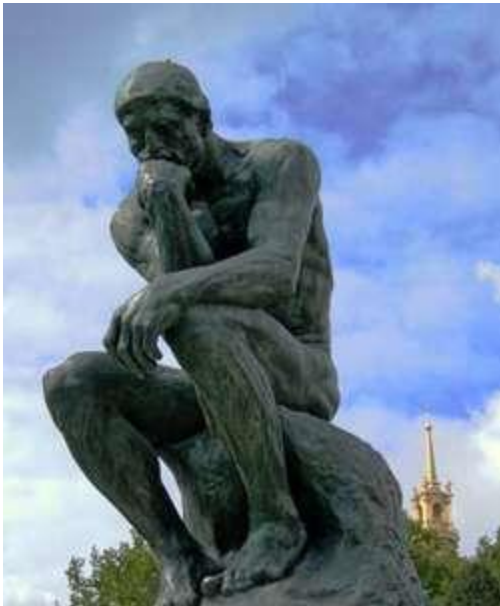
- agent's *ability to learn from experience* (to better interact with the particular environment)



Agent Properties

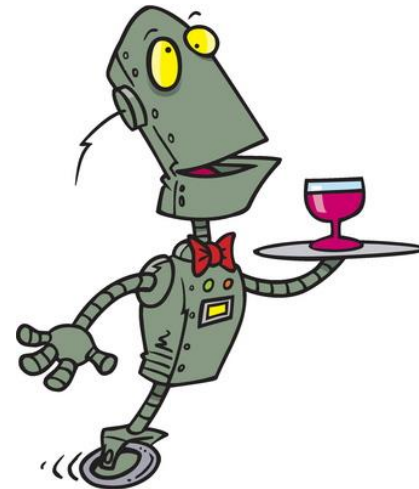
Rationality

- agent's *ability to act in a way that maximizes some utility function*



Mobility

- agent's *ability to change its location in the environment*, being it the *physical* world (robots), a *virtual* world (virtual agents), or an *electronic* network (software)



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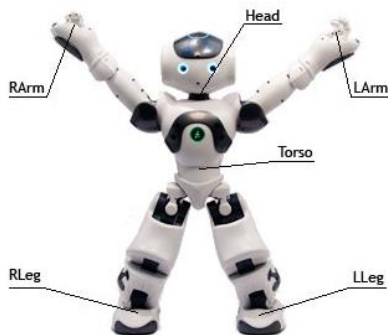
Environment

Agent systems are also characterized by
properties of the surrounding environment

Environment

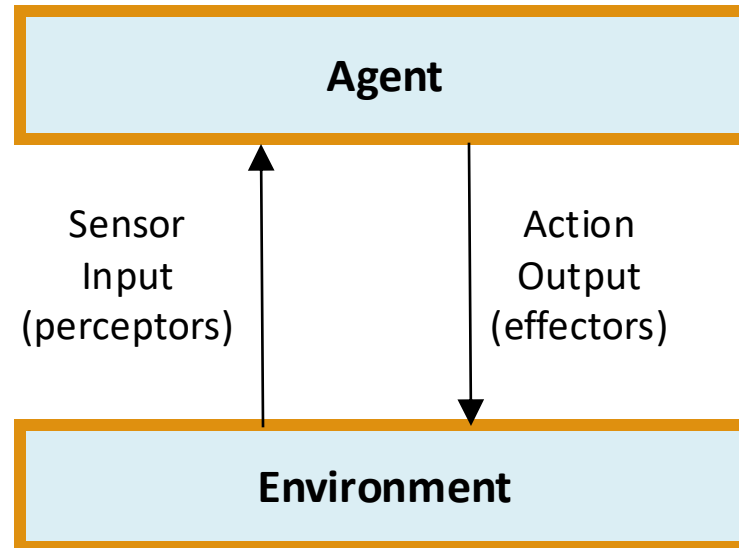
How do agents interact with the environment?

Sensors: define the *perceptors* for the agent perceive the world



Effectors: define the *actuators* for the agent perform in the world

Interaction with environment

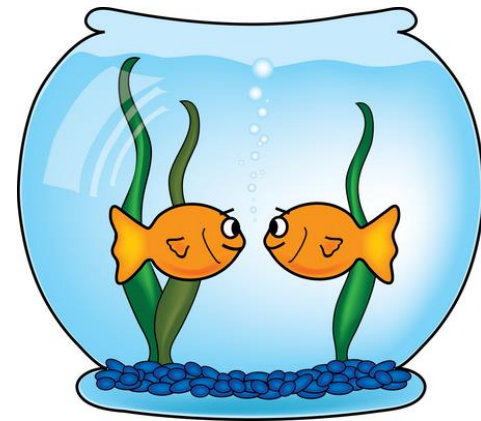


- **sensors**: cameras, speed sensor, GPS
- **actuators**: start engine, accelerate, break, turn, start light, change gears

Environment properties

Environment properties

- accessibility
- determinism
- dynamism
- continuity
- memory



Environment properties

Accessible vs. Inaccessible

- Accessible environment: *agent can obtain complete, accurate, up-to-date data about the environment's state*
- Most moderately complex environments are inaccessible



Environment properties

Deterministic vs. non-deterministic

- Deterministic environment: *action has a single guaranteed effect*
- Physical world is for us (humans) non-deterministic



Environment properties

Static vs. dynamic

- static environment: *world does not change while the agent is deliberating*



Discrete vs. continuous

- discrete environment: *fixed, finite number of possible actions and percepts*



Environment properties

Episodic vs. non-episodic

- Episodic environment: *world can be divided in a series of intervals (episodes) independent of each other*
- What happens in one episode has no influence on others



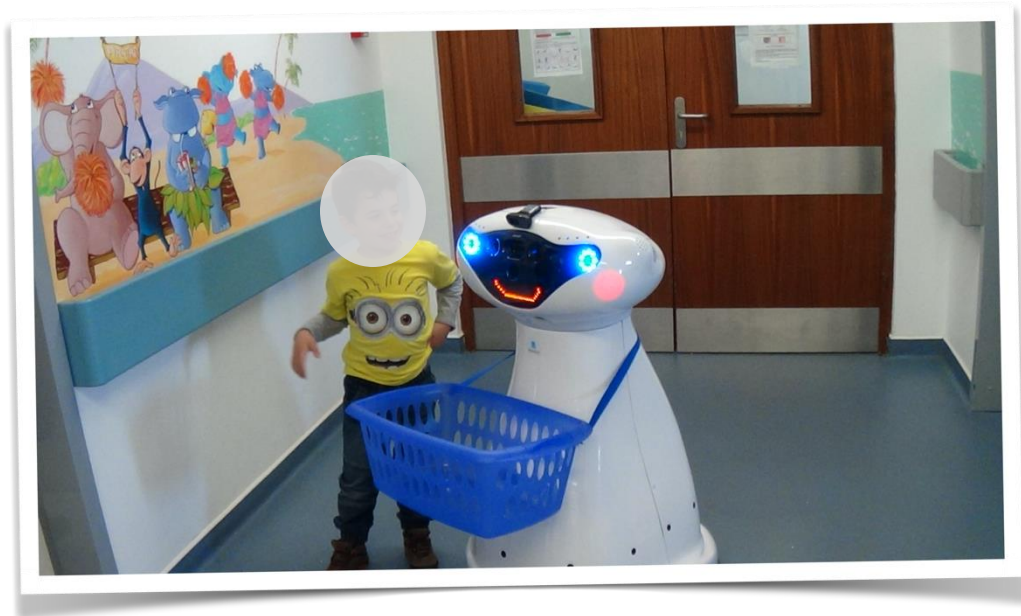
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INSIDE Project

- A **key goal** of the INESIDE project:
- Develop a networked robot system that can help kids with ASD within therapy sessions



INSIDE Project

- **Main features of the robot:**
 - **Fully autonomous** and has a key role in the therapy sessions
 - **Child-robot interaction** should be **unconstrained** as possible
 - e.g, children could freely move around the world
 - Child performs several **fun activities** in the therapy sessions



INSIDE Project

- Partners



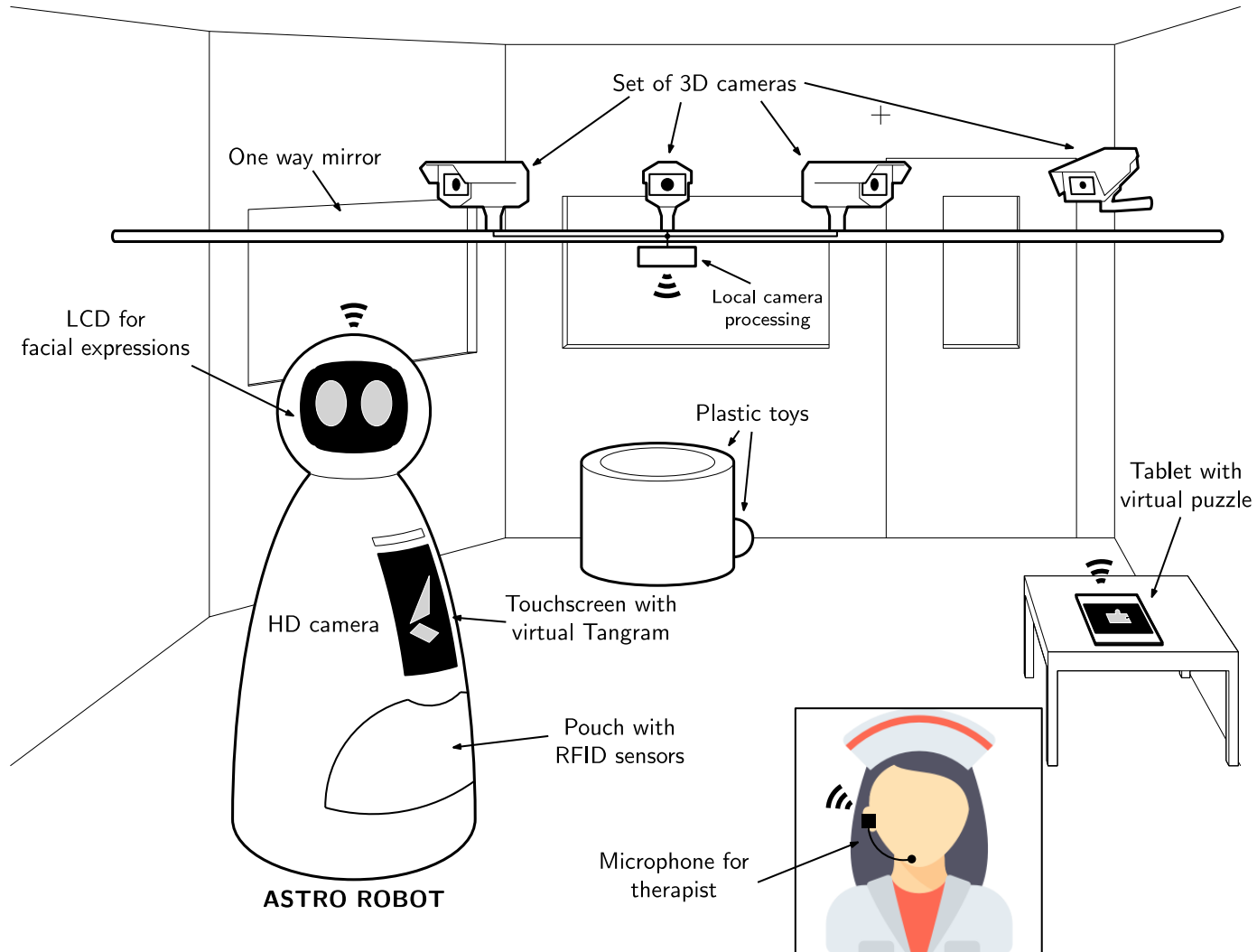
INSIDE Project

- Venue: Hospital Garcia da Horta



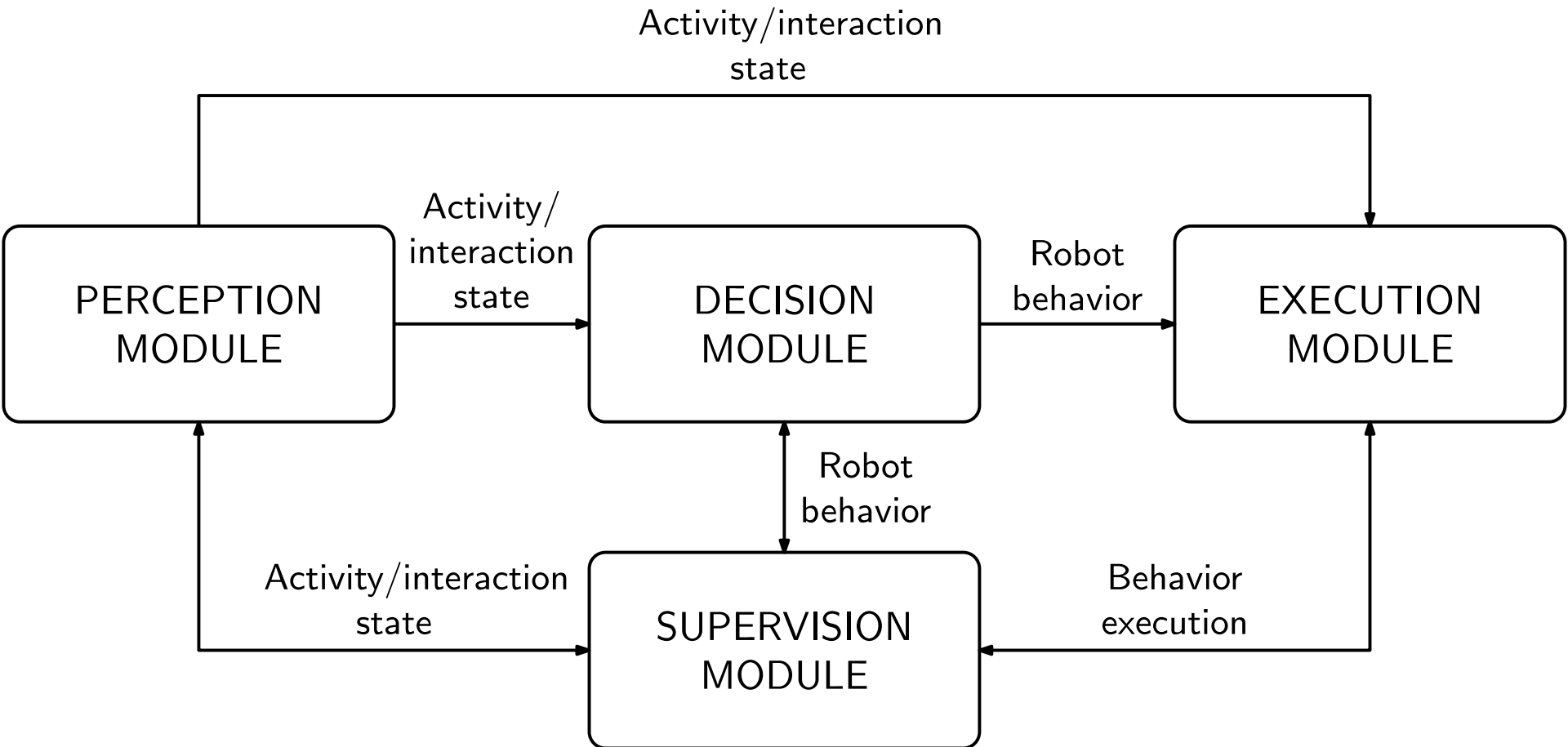
INSIDE Project

■ The INSIDE Architecture

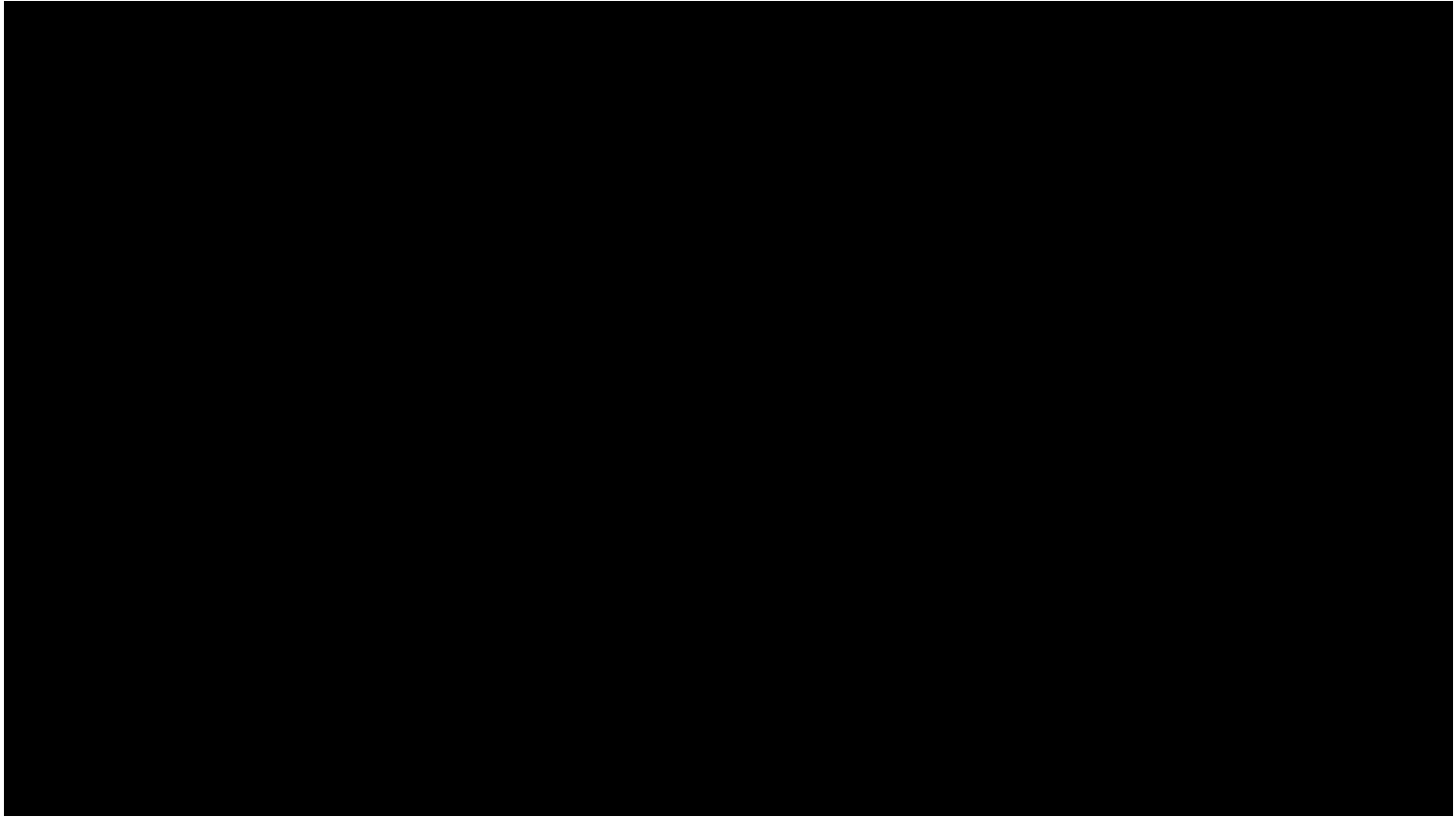


INSIDE Project

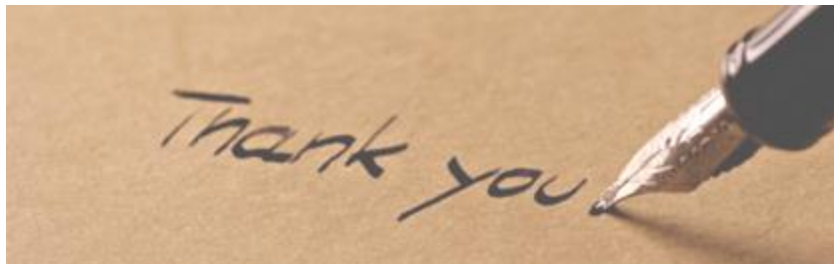
- The INSIDE Architecture



INSIDE Project



Thank You



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